Experimental verification and electromagnetic analysis of axial flux permanent magnet motor using the subdomain method

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ABSTRACT

This study investigates a subdomain method for performing the characteristic analysis of an axial flux permanent magnet motor (AFPM). Initially, the AFPM was simplified for the analytical model and analysis regions were defined to apply the subdomain method. Subsequently, a general solution was derived for the governing equations of each analysis region using the variable separation method. Subsequently, the undetermined coefficients of the general solution were obtained from the matrix using appropriate boundary conditions, and the magnetic flux density in the air-gap region was derived based on the defined magnetic vector potential. Finally, the induced voltage and electromagnetic torque of the analytical model were calculated. The analytical results were verified using the finite element method and the experimental results.

I. INTRODUCTION

An axial flux permanent magnet motor (AFPM) is used for the in-wheel and compressor of an electric vehicle because it can generate a high torque density within a limited axial length.[1] When designing an AFPM, the analytical method must be based on the appropriate assumptions and mathematical techniques. The finite element method (FEM) computes an algebraic simultaneous equation using a numerical analysis technique. Most designers use commercial software to design permanent magnet machines. Numerical analysis methods, such as FEMs have become powerful tools in electromagnetic field analysis owing to their high precision and non-linear computational capabilities; however, they are still relatively slow and time consuming. In addition, the AFPM cannot be interpreted in two dimensions because of its structure, and a three dimensional analysis is essential. In the case of 3D analysis, it takes a lot of time to confirm the analysis results according to the design variables, and it is difficult to grasp the direct physical relationship between the design variables; therefore, it is inferior to the analytical method for understanding the initial design and optimization. Analytical methods can obtain reliable results and require short analysis times. Therefore, the analytical method can be said to be suitable for AFPM design and analysis, which has difficulty in the initial design and optimization owing to 3D analysis.[2,5]

Therefore, this study presents a method for deriving the electromagnetic characteristics of an AFPM using the subdomain method. First, the analysis model was simplified, and the governing equations and general solutions were derived through appropriate assumptions and separation of variables. In addition, the magnetic flux density was derived using the matrix through the boundary conditions in each analysis region, and the induced voltage and electromagnetic torque were derived. The results of the subdomain method were verified using the FEM and experimental results.

II. ELECTROMAGNETIC ANALYSIS OF AN AXIAL FLUX PERMANENT MAGNET MOTOR

A. Simplified analytical model

The topology of an AFPM is ideal for designing a modular machine in which the number of identical modules can be adjusted according to the power and torque requirements.[3] That is a double-sided rotor can produce twice the output of a single-sided rotor, and this characteristic is suitable for designing simplified analytical models. Figures 1(a) and 1(b) depict the analysis model and simplified analytical model, respectively. z_1, z_2, z_3, and z_4 denote the heights from the rotor yoke, surface of the permanent magnet (PM), inner side of the stator, and height of the slots, respectively. β represents the angle of the slot, and θ_i represents the angular position of the ith slot and can be expressed as

$$\theta_i = -\frac{\beta}{2} + \frac{2\pi}{Q} i, \text{ with } 1 \le i \le Q \tag{1}$$

where Q denotes the number of slots. The analytical model comprises three regions: a PM region (Region I), an air gap region (Region II), and an ith slot region (Region i). The main variables used in the analytical method are listed in Table I.

TABLE I. Main design variables of AFPM.

Variables	Description	Value
P	Number of poles	14
Q	Number of slots	12
n	Number of harmonics about PM, air gap region	150
k	Number of harmonics about slot region	20
β	Angle of the slot	18°
N_{turns}	Number of turns	32
r	Central value between the inner and outer diameters of the rotor	39.25 mm
z_1	Height of the rotor yoke	3 mm
z_2	Height to PM	6 mm
z_3	Height to inside of stator	7 mm
z_4	Height to slot	18.5 mm

B. Magnetic field characteristics of AFPM

The three regions exhibit a symmetric structure with respect to the z-direction and are expressed in the coordinate system. The magnetic vector potential of each region is expressed as follows:

$$\mathbf{A}^{\mathbf{I}}_{\mathbf{rn}} = A^{I}_{rn}(z,\theta)\mathbf{i}_r \quad \mathbf{A}^{\mathbf{II}}_{\mathbf{rn}} = A^{II}_{rn}(z,\theta)\mathbf{i}_r \quad \mathbf{A}^{\mathbf{i}}_{\mathbf{rk}} = A^{i}_{rk}(z,\theta)\mathbf{i}_r \tag{2}$$

where n and k denote the spatial harmonics in each region. The governing equations can be derived from Maxwell's equations. The absence of a free current in the PM region implies $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ and $\nabla \times \mathbf{H} = 0$, resulting in the derivation of $\mathbf{B} = \mu_0\mathbf{M}$. The magnetic vector potential $\mathbf{A}$ is defined to obtain $\nabla \times \mathbf{A} = \mathbf{B}$.[4] Because of the absence of magnetization of the free current in the air gap, the governing equation for Region II can be expressed using the Laplace equation. Furthermore, as $\mathbf{M} = 0$ and $\nabla \times \mathbf{H} = \mathbf{J}$ owing to the free current generated by the winding and the absence of a magnet in the slot, the governing equation can be expressed using the Poisson equation. Thus, the form of the governing equations for each region is as follows:

$$\frac{1}{r^2}\frac{\partial A^{I}_{rn}}{\partial \theta^2} + \frac{\partial A^{I}_{rn}}{\partial z^2} = -\mu_0(\nabla \times \mathbf{M}) \tag{3}$$

$$\frac{1}{r^2}\frac{\partial A^{II}_{rn}}{\partial \theta^2} + \frac{\partial A^{II}_{rn}}{\partial z^2} = 0 \tag{4}$$

$$\frac{1}{r^2}\frac{\partial A^{i}_{rk}}{\partial \theta^2} + \frac{\partial A^{i}_{rk}}{\partial z^2} = -\mu_0\mathbf{J} \tag{5}$$

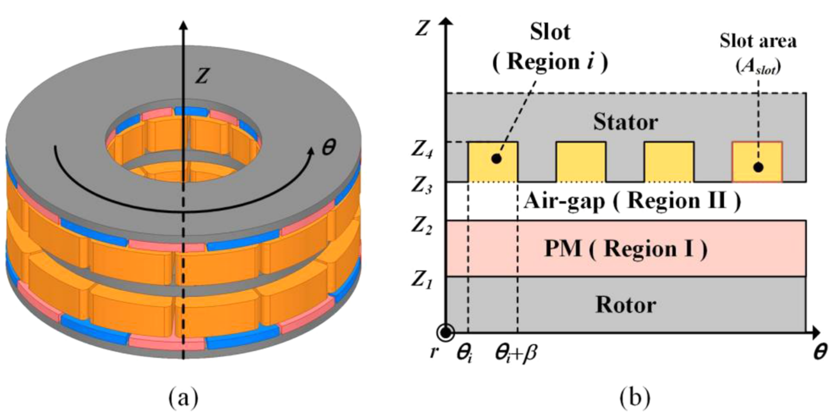

FIG. 1. Axial flux permanent magnet motor: (a) analysis model, and (b) simplified analytical model.

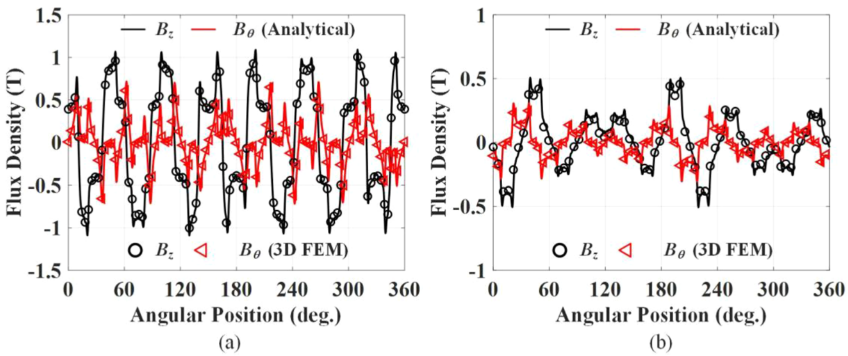

FIG. 2. Comparison of (a) magnetic flux density and (b) armature reaction analysis result.

General solutions to Eq. (2) can be derived using the variable separation method. In addition, because the magnetic vector potential $\mathbf{A}$ comprises variables r and θ, it can be expressed as $\mathbf{A} = Z(z)\Theta(\theta)\mathbf{i}_z$. By substituting this into Eq. (2), a general solution can be derived for each region.

$$\mathbf{A}_{\mathbf{rn}}^{\mathbf{I}} = A_0^I + B_0^I z + \sum_{n=1}^{\infty} \times \left[\begin{array}{l} \left(A_n^I e^{\frac{n}{r}z} + B_n^I e^{-\frac{n}{r}z} - \frac{\mu_0 r M_{zn}}{n} \cos(n\theta_0) \right) \sin(n\theta) \\ + \left(C_n^I e^{\frac{n}{r}z} + D_n^I e^{-\frac{n}{r}z} + \frac{\mu_0 r M_{zn}}{n} \sin(n\theta_0) \right) \cos(n\theta) \end{array} \right] \mathbf{i}_r \quad (6)$$

$$\mathbf{A}_{\mathbf{rn}}^{\mathbf{II}} = A_0^{II} + B_0^{II} z + \sum_{n=1}^{\infty} \left[\left(A_n^{II} e^{\frac{n}{r}z} + B_n^{II} e^{-\frac{n}{r}z} \right) \sin(n\theta) + \left(C_n^{II} e^{\frac{n}{r}z} + D_n^{II} e^{-\frac{n}{r}z} \right) \cos(n\theta) \right] \mathbf{i}_r \quad (7)$$

$$\mathbf{A}_{\mathbf{rk}}^{\mathbf{i}} = A_0^i + B_0^i z - \frac{\mu_0 J_0^j}{2} z^2 + \sum_{k=1}^{\infty} \left[\left(A_k^i e^{\frac{k\pi}{r\beta}z} + B_k^i e^{-\frac{k\pi}{r\beta}z} - \frac{\mu_0 J_k^i}{2 - \left(\frac{zk\pi}{r\beta}\right)^2} z^2 \right) \cos\left(\frac{k\pi}{\beta}(\theta - \theta_i) \right) \right] \mathbf{i}_r \quad (8)$$

where θ_0 is the initial position of the rotor; ${J_0}^i$ and ${J_k}^i$ are the current density components; and r is the central value between the inner and outer diameters of the rotor. ${A_0}^{\mathrm{I,II},i}$, ${B_0}^{\mathrm{I,II},i}$, ${A_n}^{\mathrm{I,II}}$, ${B_n}^{\mathrm{I,II}}$, ${C_n}^{\mathrm{I,II}}$, ${D_n}^{\mathrm{I,II}}$, ${A_k}^i$, and ${B_k}^i$ represent undetermined coefficients. The magnetic flux density can be expressed through the derived magnetic vector potential, as shown in Eqs. (6)–(8). Therefore, it can be expressed as Eq. (9).

$$\mathbf{B_z} = -\frac{1}{r}\frac{\partial \mathbf{A}}{\partial \theta}\mathbf{i}_z \quad \mathbf{B_\theta} = \frac{\partial \mathbf{A}}{\partial z}\mathbf{i}_\theta \quad (9)$$

As shown in Fig. 2, the computed magnetic flux density and armature reaction results from the analysis model can be verified by comparison with the 3D FEM results.

C. Performance analysis of the AFPM

The induced voltage is proportional to the temporal variations in the flux linkage and is induced in a direction that opposes changes within the flux linkage.[6] Therefore, accurate derivation of the flux linkage is important for predicting the induced voltage. The expression for the flux linkage is shown in Eq. (10), which can be used to derive the induced voltage by applying Faraday's law.

$$\lambda = \frac{N_{turns}}{2A_{slot}}(R_o - R_i)^2 \int_{z_3}^{z_4} \int_{\theta_i}^{\theta_i+\beta} A_{rk}^i d\theta dz \quad (10)$$

where N_{turns} represents the number of turns in the coil; A_{slot} is the slot area; and R_o and R_i denote the outer and inner diameters of the rotor, respectively. Based on Stoke's theorem, the line integral of the magnetic vector potential can be used to express the magnetic flux. The average value of the magnetic vector potential can be obtained by dividing the surface integral calculation of the magnetic vector potential for the slot region by the slot area. In addition, electromagnetic torque can be derived using the Maxwell stress tensor. According to the Maxwell stress tensor, electromagnetic torque is influenced by the magnetic flux density.[7] Therefore, the torque equation can be expressed as shown in Eq. (11).

$$T_e = \frac{\left(R_o^3 - R_i^3\right)}{3\mu_0} \int_0^{2\pi} B_z^{air}(z,\theta) B_\theta^{air}(z,\theta) d\theta \quad (11)$$

III. RESULTS AND DISCUSSION

Figures 3(a) and 3(b) present the results of the calculated flux linkage and phase-induced voltage, respectively, compared with those of the 3D FEM. Compared with the 3D FEM analysis, the flux linkage showed an error of 0.39%, whereas the phase-induced voltage exhibited an error of 3.51%. Figure 4(a) depicts the experimental equipment setup for the AFPM, while (b) shows the stator and rotor of the actual model.

Figure 5(a) displays the line-to-line induced voltage compared with the 3D FEM, and (b) compared the peak to peak value of the line-to-line induced voltage with the experimental and analysis results. Additionally, (c) shows the electromagnetic torque, indicating the average values obtained through the analytical, 3D FE, and measurement methods.

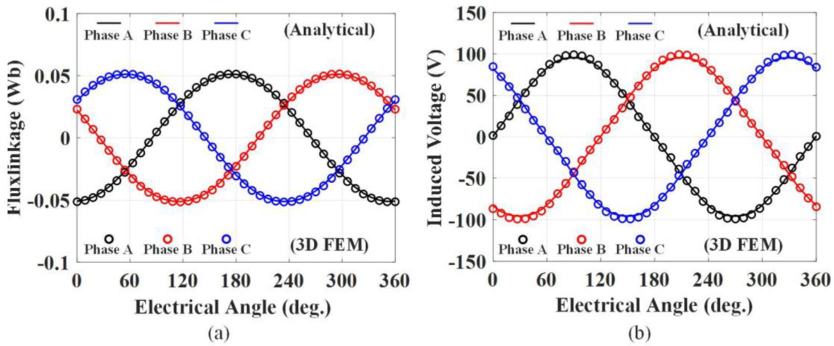

FIG. 3. Analysis results of analytical method vs. 3D FEM: (a) flux linkage and (b) phase induced voltage.

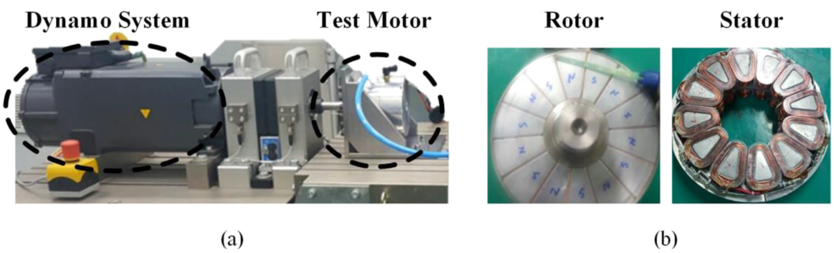

FIG. 4. (a) Experimental set, and (b) actual manufactured rotor and stator.

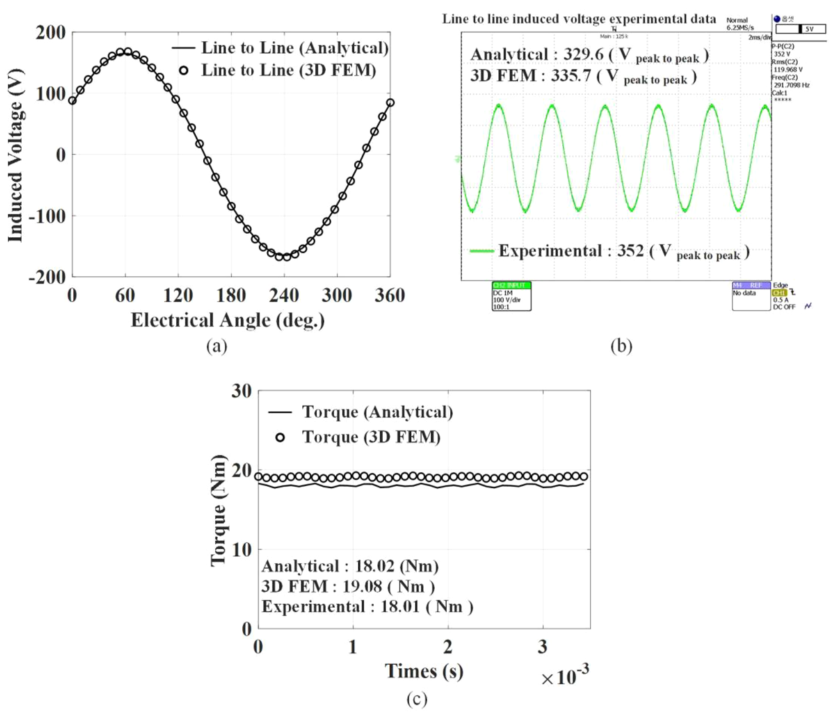

FIG. 5. (a) Line to line induced voltage results of the analysis model, (b) line to line induced voltage experiment results, and (c) comparison of analytical, 3D FE and experiment results for electromagnetic torque.

IV. CONCLUSION

This study presents a characteristic analysis of the AFPM using the subdomain method. The magnetic flux density, induced voltage, and electromagnetic torque were derived using the subdomain method. All results were verified using the finite element method and experimental results. The AFPM requires 3D analysis; however, approximating the analytical model to 2D allows for a quick assessment of its initial electromagnetic characteristics. Therefore, the subdomain method proposed in this study can be considered useful for the initial design of AFPMs.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jun-Won Yang: Writing – original draft (equal). **Hoon-Ki Lee**: Formal analysis (equal). **Su-Min Kim**: Formal analysis (equal). **Woo-Sung Jung**: Formal analysis (equal). **Yong-Joo Kim**: Funding acquisition (equal). **Kyung-Hun Shin**: Writing – review & editing (equal). **Jang-Young Choi**: Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available on request from the corresponding authors. The data are not publicly available due [state restrictions such as privacy or ethical restrictions].

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